Isogeny volcano cost, PROTOCOL v2 / v2.1: the ℓ = 2 IC-yield lead is confirmed

Question. v1 (`../isogeny-volcano-cost_20261017`) left one exploratory lead: on ℓ = 2 volcanoes, floor curves gave about 6.5% fewer index-calculus relations per trial than crater curves, while ℓ = 3 volcanoes showed nothing. Does this hold on fresh volcanoes with many more crater curves? Are rho and IC yield equal across levels where v1 could not say?

The protocol was frozen before any data (`PROTOCOL.md`):

- **v2:** 8 classes.
- **Amendment v2.1:** 20 more classes, added before any v2 record was analysed.

The headline is v2.1, with all 28 classes. The original 8 are reported beside it.

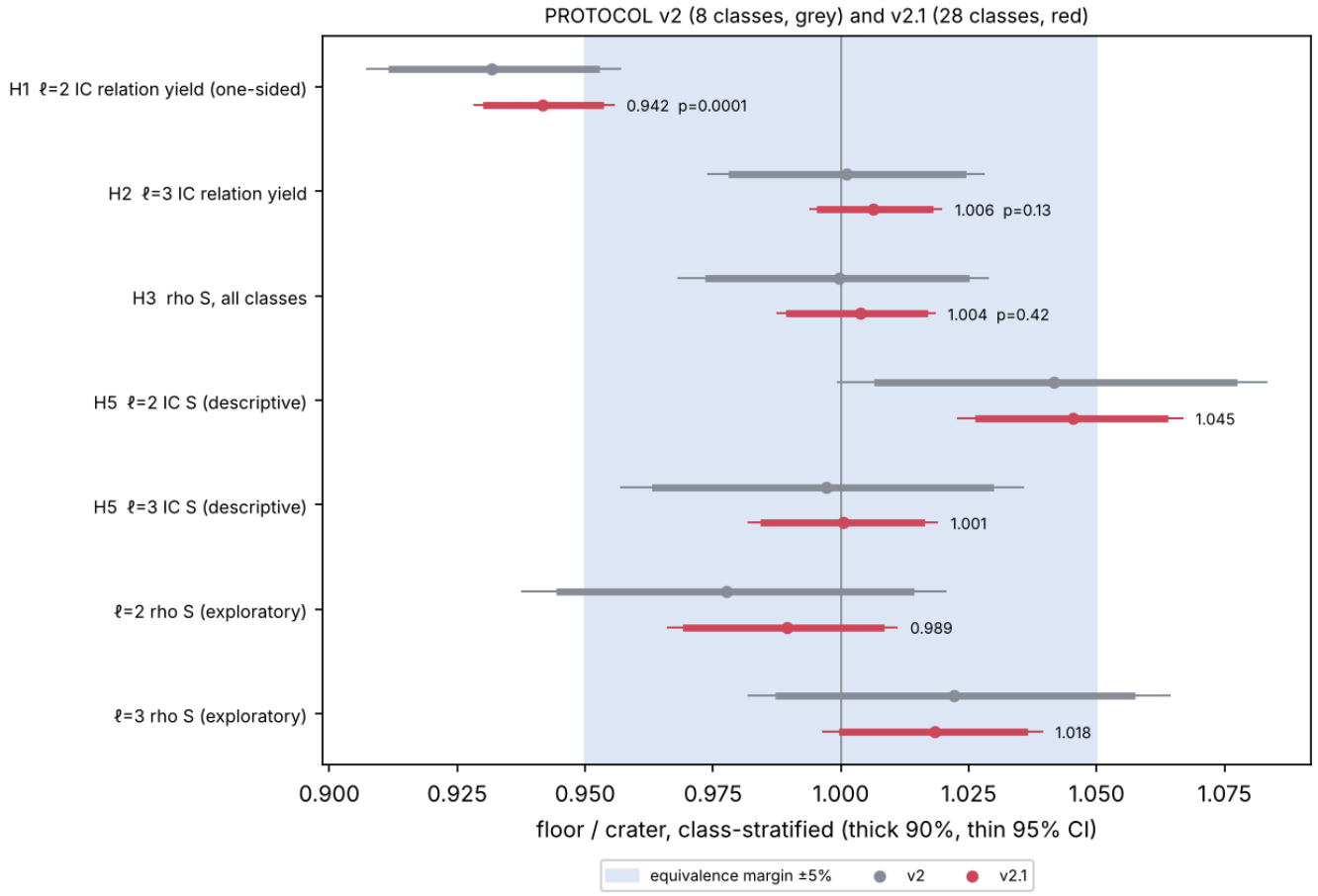
Answer

Measured:

- 28 classes (14 with ℓ = 2, 14 with ℓ = 3), 24-bit `p`, depth-3 volcanoes with split craters.
- 7,760 curve-models: every crater curve in 16 isomorphic models, lower levels in 4.
- 154,160 measured runs (each after one warm-up), all verified.
- 113 sessions audited; 1,344 of 1,344 replays exact.

hypothesis	v2.1 (28 classes): floor/crater, 90% CI [95% CI]	p	v2 (8 classes)	verdict
H1 (primary): ℓ = 2 IC relation yield is lower on the floor	**0.942** [0.931, 0.953] [0.928, 0.956]	0.0001 (one-sided, 10,000 perms)	0.932 [0.912, 0.952], p = 0.0001	**confirmed**: ℓ = 2 floors give 5.8% [4.4, 7.2] fewer IC relations per trial than craters
H2: ℓ = 3 IC relation yield is equal across levels	1.006 [0.996, 1.017]	0.13	1.001 [0.979, 1.024]	**equal within ±5%**
H3: rho cost is equal across levels	1.004 [0.990, 1.016]	0.42	1.000 [0.974, 1.024]	**equal within ±5%**
H4: the v1 coordinate effect replicates (IC yield)	model τ 7.2% (null p95 4.7%), Q p = 0.0005; horizontal τ 1.9%		model τ 7.2% > horizontal 2.3%, p = 0.0005	**replicated**
H5 (descriptive): IC cost S, floor/crater	ℓ = 2: 1.045 [1.023, 1.067]; ℓ = 3: 1.001 [0.982, 1.019]		ℓ = 2: 1.042; ℓ = 3: 0.997	ℓ = 2 floors cost 4.5% more S

Every curve-model's ratios are in `analysis/`. H1 is significant under Holm over H1–H3: its p = 0.0001 is below the smallest threshold, 0.0167. H2 and H3 are not significant differences, and each 90% CI lies inside [0.95, 1.05]. v2 and v2.1 agree on every row.



Positive controls (all required ones pass)

control	required for	result
Planted IC (factor base 33 against 32, yield matched by workload; predicted 1.063)	H1, H2	ℓ = 2: 1.061 [1.058, 1.064]; ℓ = 3: 1.063 [1.060, 1.065]; all: 1.062 [1.060, 1.064]
BSGS exactness	all	1 distinct cost per paired target on all 56 targets (28 classes × 2)
H1 power: `--plant-level 3:0.95` on the ℓ = 3 IC yield, one-sided	H1	0.956, p = 0.0001 < 0.0167 (v2: 0.951, p = 0.0001)
H3 power: `--plant-level 3:1.05` on rho, all classes	H3	1.054, p = 0.0001 < 0.0167 (v2: 1.050, p = 0.0001)
Single curve ×1.15 (`l2-s1001`, crater vertex 2), max|z|	reported only	IC yield: detected (max|z| p = 0.019). Rho: **not detected** (p = 0.42)

The single-curve control is a limit on what the study can see. A 15% rho shift on one curve among about 30 is below what the max|z| test detects at this N.

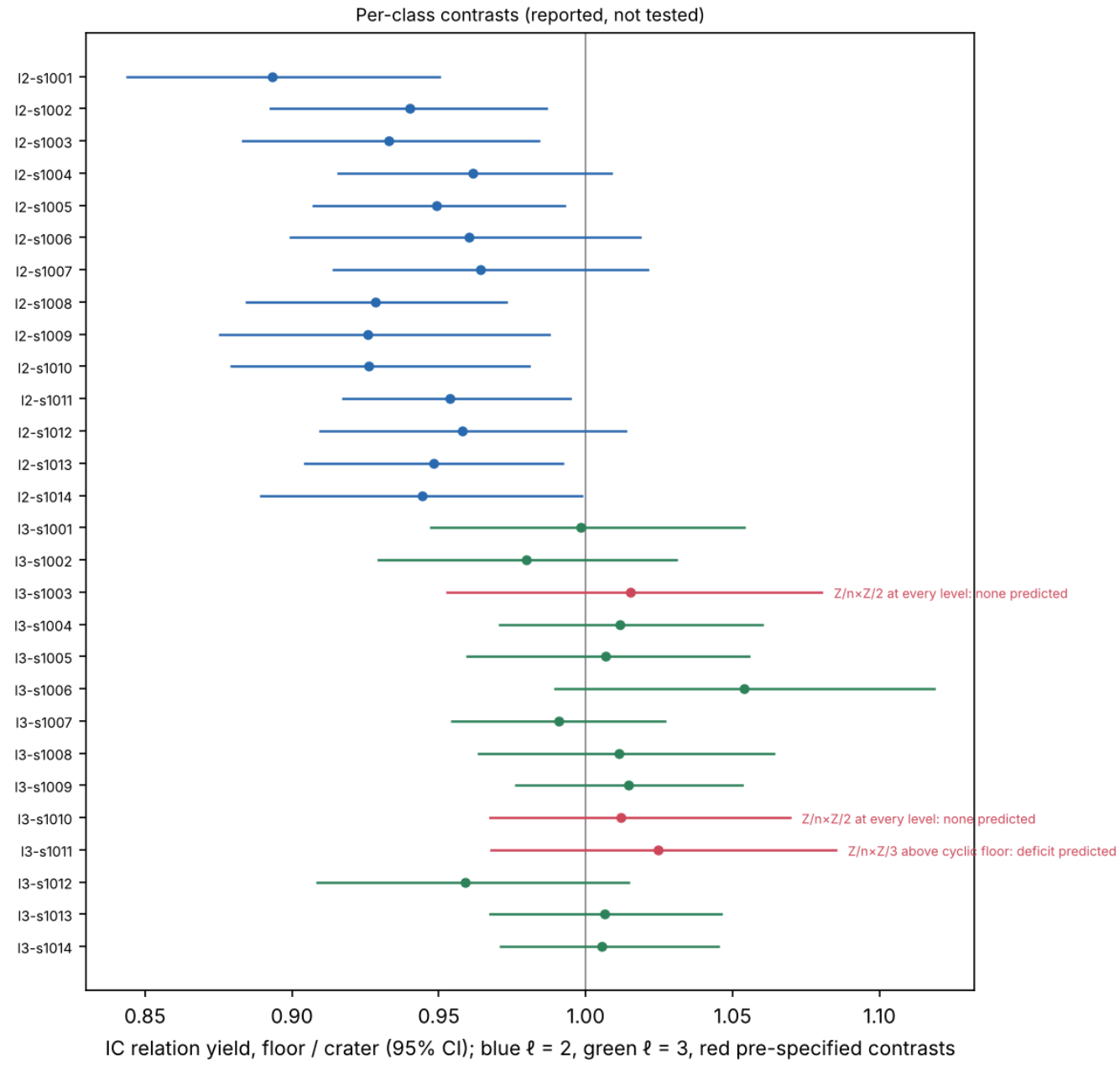
Mechanism contrasts (pre-specified, reported, not tested)

The amendment predicted that if the ℓ = 2 deficit comes from the ℓ-torsion structure changing between levels, then:

- `l3-s1011` (Z/n × Z/3 above a cyclic floor) would show a floor deficit;
- `l3-s1003` and `l3-s1010` (Z/n × Z/2 at every level) would show none.

class	IC yield floor/crater [95% CI]	prediction
`l3-s1011`	**1.025 [0.968, 1.085]**	deficit: **not seen**
`l3-s1003`	1.015 [0.953, 1.080]	none: as predicted
`l3-s1010`	1.012 [0.967, 1.069]	none: as predicted
every ℓ = 2 class (14)	0.893 to 0.964: every estimate below 1, and every per-class permutation p < 0.05; 10 of 14 have a 95% CI wholly below 1	deficit in every class

The deficit is uniform across ℓ = 2 volcanoes. It does not appear in the one ℓ = 3 volcano with the analogous structure change. So the explanation "the ℓ-power torsion changes between levels" is not supported, and the cause of the ℓ = 2 effect stays open. One class is weak evidence (its CI reaches 0.968), so this is reported as a contrast, not a rejection.



Exploratory (not pre-registered)

- **Rho split by ℓ:**
  - ℓ = 2: 0.989 [0.970, 1.008] (90%).
  - ℓ = 3: 1.018 [1.000, 1.036] (90%), permutation p = 0.0054.

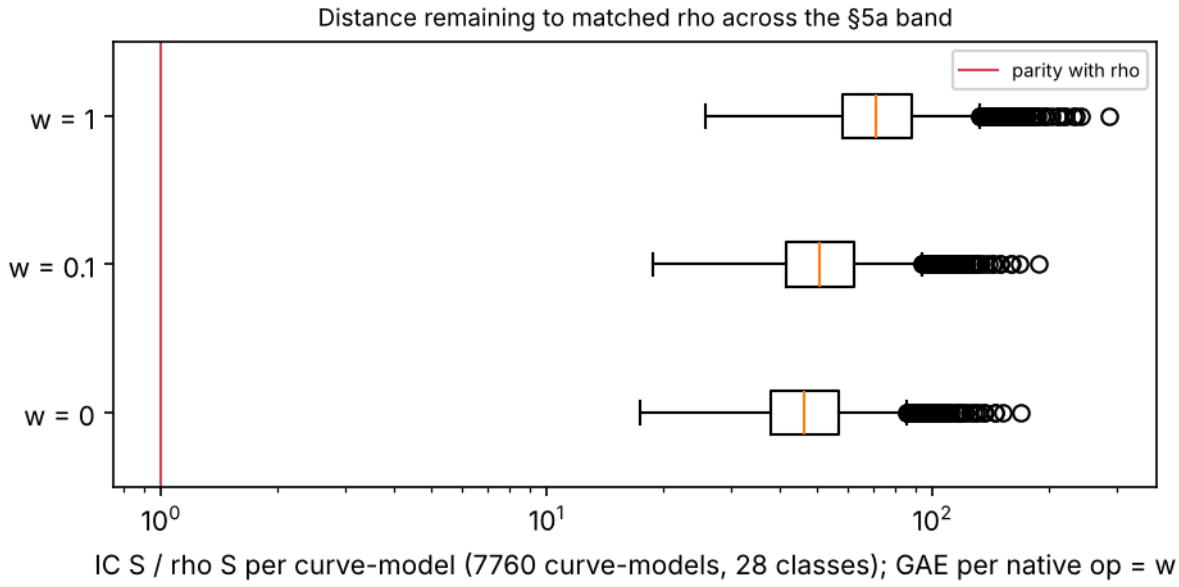
  Both are inside ±5%, and they cancel in H3. The ℓ = 3 sign is a lead at most.
- **Horizontal effect:** IC yield differs between same-level curves, τ = 1.9% (p = 0.0005). That is a quarter of the coordinate effect (7.2%). Rho shows no horizontal effect beyond the null (τ 1.9%, null p95 1.9%, p = 0.07).

Gap to rho (CLAUDE.md success criterion; AGENTS.md §5a)

`ecbench band` on every class's main session, at weights 0, 0.1 and 1 GAE per native operation: **rho is below IC on all 7,760 curve-models at every weight.**

weight	IC S / rho S, min	median	max
0	17.4	46.5	169.5
0.1	18.9	50.8	188.2

1	25.8	71.4	286.8



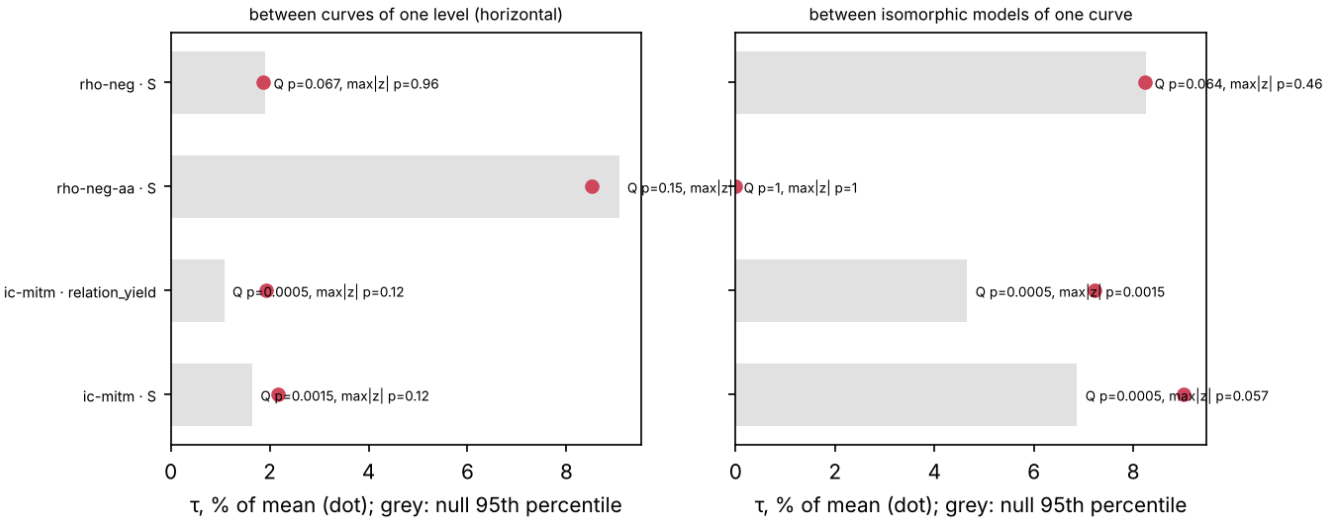
Is anything here an improvement over the IC baseline? H1 and H5 give a measured, pre-registered difference: on an $\ell = 2$ floor curve, IC costs 4.5% [2.3, 6.7] more S than on that volcano's crater curves. A floor-curve target could therefore walk up the volcano (an isogeny walk of depth 3) and run IC on a crater curve.

This is recorded as a **candidate, not a claimed improvement**:

- the isogeny walk's cost is not charged in these sessions;
- it is 24-bit only;
- its mechanism is unknown.

The gap to rho it would leave is still 17–170× at weight 0.

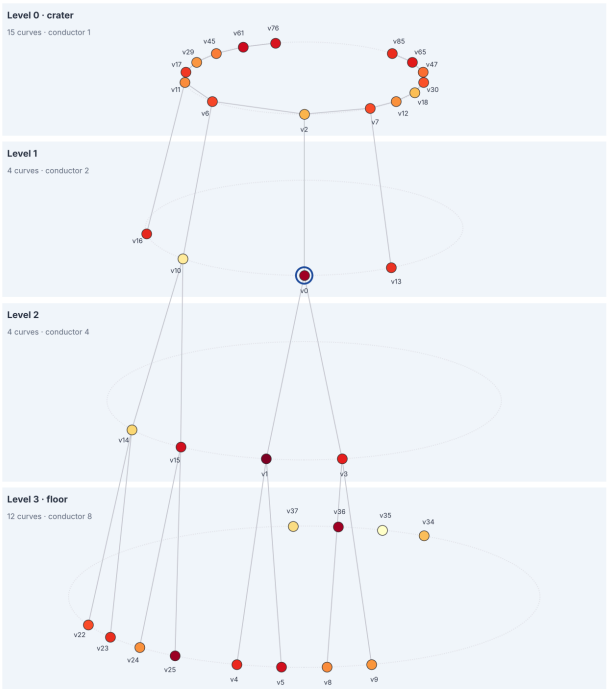
Heterogeneity and the volcanoes

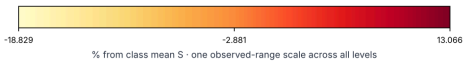


There is one heat map per class in `figures/volcano-<class>.png`, made from that class's own analysis (`analysis/class-<class>.json`). Two examples follow: the $\ell = 2$ class with the largest deficit, and the $\ell = 3$ contrast class.

I2-s1001: 2-isogeny volcano over F_15808601 · crater curves × 16 models, lower levels × 4, 8 paired targets
PROTOCOL v2.1 (research/isogeny-volcano-cost-v2_20261017) · % from the class mean · ringed blue: v0 = root

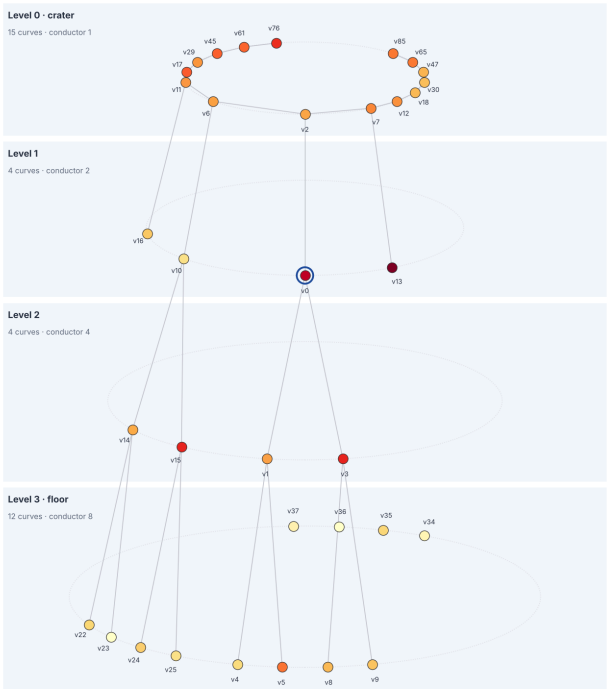
rho.negation · mean S

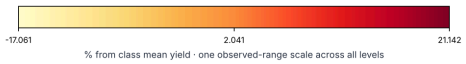




% from class mean S · one observed-range scale across all levels

ic.pipeline mitm · relation yield



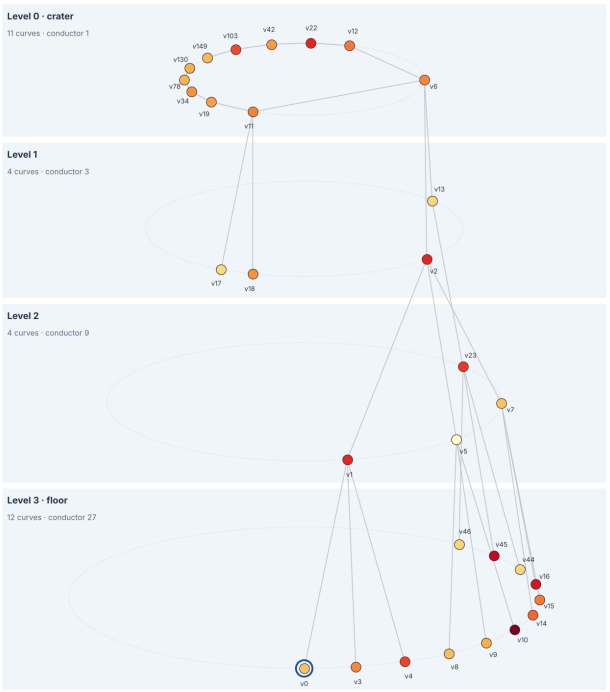


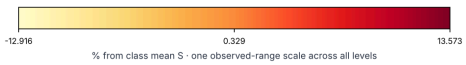
% from class mean yield · one observed-range scale across all levels

Colours span each panel's observed range. Read them against the per-curve CIs and the horizontal test in analysis/class-<name>.json before reading a pattern.

I3-s1011: 3-isogeny volcano over F_11719327 · crater curves × 16 models, lower levels × 4, 8 paired targets
PROTOCOL v2.1 (research/isogeny-volcano-cost-v2_20261017) · % from the class mean · ringed blue: v0 = root

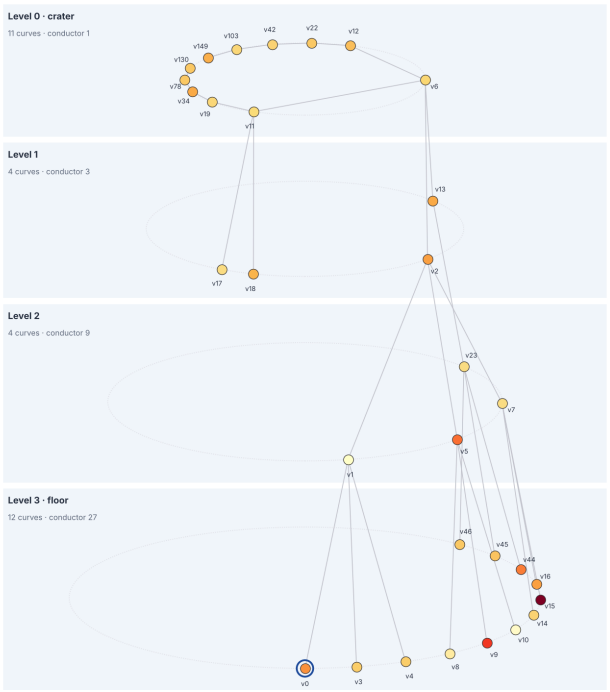
rho.negation · mean S

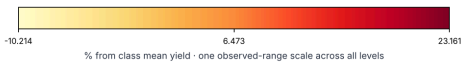




% from class mean S · one observed-range scale across all levels

ic.pipeline mitm · relation yield





% from class mean yield · one observed-range scale across all levels

Colours span each panel's observed range. Read them against the per-curve CIs and the horizontal test in analysis/class-<name>.json before reading a pattern.

Evidence and disclosures

- **Specs:** `classes/<class>/spec*.json`, `curves.jsonl`, and the export logs `classes/<class>.log`. The digests are in PROTOCOL.md: v2 `0f7c6376…`, all 28 classes `ed9a7b88…`. They were computed from the repository root, with full paths. Re-run that way on this commit, the 28-class digest reproduces `ed9a7b88…` exactly, so no spec changed after freezing:

  ```
  D=research/isogeny-volcano-cost-v2_20261017
  sha256sum $D/classes/*/spec*.json $D/classes/*/curves.jsonl | sha256sum
  ```

- **Sessions:** `sessions/<class>/<spec>/`, with `records.jsonl` and `plan.json` gzipped. `sessions/MANIFEST.sha256` holds the SHA-256 of every uncompressed file. The CI audit (`.github/scripts/ecbench-audit-sessions.sh`) verifies a decompressed copy.
- **Audits:** `audits/<class>__<spec>.json` from `ecbench verify --replay 12`. All 113 are `ok`, with 1,344/1,344 replays reproduced.
  - The audits count 308,511 records: 154,160 measured runs, as many warm-ups, and the 191 runs of the interrupted session below.
- **Analysis:** `analyse.sh` runs every `volcano-stats` and `ecbench band` call behind every number in this README:
  - stratified vertical statistic (v1.1);
  - bootstrap 2,000;
  - 10,000 permutations; 2,000 for the per-class runs;
  - seed 20261017.

  The figures come from `plot.py`, which only draws recorded values.
- **Interrupted session:** `l2-s1002/spec` was stopped at 191 runs, because a second stream claimed the same class. It is kept as evidence (audit `ok`, `--allow-interrupted`). Every run is claimed atomically, so no run was duplicated. The class was re-run whole as `spec-r2`, and the analysis uses `spec-r2`.
- **Parallel streams:** the sessions ran as up to four parallel streams, each with its own `--lock` (`/tmp/crypto-bench-g1..g4.lock`). This is not the global bench lock, so the streams shared the 4 vCPUs.
  - Only counted units (S, relations, trials) are cited; no wall time is.
  - The host is L0: a shared 4-vCPU VM (Intel Xeon @ 2.80 GHz).
- **Operations:** the disk filled once while sessions ran (a full git worktree). No session was affected: every audit is `ok`. The groups of 5 classes in the amendment were run as single-class jobs, to stay under the host's 2-hour background limit.
- **Scope:** 24-bit `p`, depth 3, split craters, $\ell \in \{2, 3\}$, and the three methods at v1's parameters. No security statement is made, and no independent-host replay was run.

The report PDF is `report.pdf`. The figures and PDF are also in `benchmarks/20261017-isogeny-volcano-cost-v2/`. Its row in `benchmarks/README.md` is admitted under the validation gate (AGENTS.md; `docs/VALIDATION_GATE.md`) by an independent validator's `VALIDATION.json`. The validator:

- replayed 583 of 583 runs, drawn from a nonce of its own;
- re-checked every scalar with its own curve arithmetic;
- recomputed the headline numbers from the raw records with its own program (`validation/vrecompute.rs`).

The rho reference is `rho.negation` in the same sessions and binary.